

The best answer to your storage woes is to share the load

Divided We Conquer

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About as mystical and “ooh”-provoking as the Matrix itself is this crazy new concept called The Grid. Still being researched by universities and big corporations alike, Grid Computing will change the way the network works. Soon, we will be able to harness the processing power of every computer on the Grid to make complex processing the matter of a jiffy. This, however, is a scenario we’ll have to wait for—for now.

And then there’s the case for Grid Storage. The term is quite suggestive—grid storage is when you don’t have one dedicated file server for your LAN; rather, the task of storing data can be extended to nearly every computer on your network.

The Grid And You

So what use could you possibly derive from a storage grid in your office? Well, for one thing, you won’t need to invest in a powerful built-for-serving-files machine as a file server—you could actually make do with a few half-decent PCs that nobody is using any more.

More important, though, is the question of failure. When your central file server decides to pack up, you and your team will most likely spend the rest of the day twiddling your thumbs or playing table-tennis, generally not doing much to earn their worth—we know from experience! However, a storage grid is quite like having a RAID (Redundant Array of Independent Devices) setup over a LAN—if one unit goes poof, there are always the others to keep your data available.

Still not convinced? With a storage grid, you can squeeze the most out of old hardware even after it’s time to upgrade the rest of your PCs—just add it to the grid and go!

By now, you’re probably curious to know how exactly you can turn your office LAN into a storage grid, so here goes.

WebDAV

WebDAV stands for Web-based Distributed Authoring and Versioning. When Tim Berners-Lee conceptualised the Internet, one of his intentions was to make it a writable medium as well—basically, you should be able to use the Internet the way you would use your own hard disk. The WebDAV protocol is an extension of the HTTP protocol for data transfer over the Web. Nearly all operating systems come with inherent support for WebDAV, letting you use files on the Internet as if they were stored right on your PC.

GridBlocks DISK

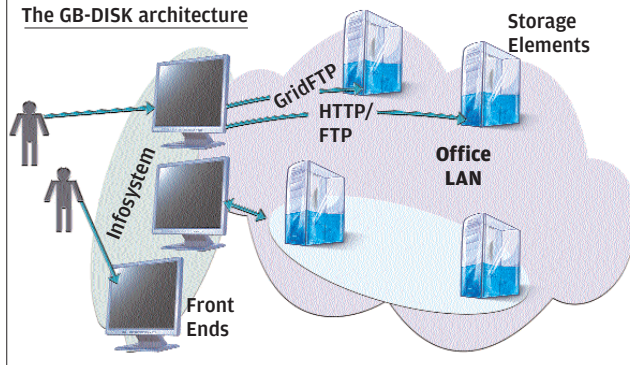
GridBlocks is an open source project by the Helsinki Institute of Physics. GridBlocks DISK (Distributed Inexpensive Storage with K-Availability) is one of the three applications they are working on—its purpose is quite obvious from the acronym. Unlike solutions such as SCSI or SATA RAIDs that would set you back a pretty penny, GridBlocks DISK (GB-DISK) will make full use of the resources you give it, with a near-zero investment.

GB-DISK ships as Java .jar files, and is thus completely platform-independent. So even if your grid is a mix of Windows and Linux servers, you should face no problem at all. If you wish to try out GB-DISK, you can find it on this month’s DVD.

How GB-DISK Works

The GB-DISK architecture consists of two main components—the Front End service (FE) and the Storage Element service (SE). Users will interact with the FE, which will break the file into “stripes” and spread them to SEs on the network.

The GB-DISK architecture



It also saves the metadata—data about the files—to the Infosystem, where it is accessible to all GB-DISK services on the network.

Downloading files from the grid works in the opposite way. When a user requests a file using the FE, it uses the metadata to determine where it has stored the different fragments of the file, and begins to download them. Once all the fragments have arrived, the FE strings them together to form the original file.

All this isn’t apparent to the user—all he or she will see is a plain old Web page or Web folder.

Firing It Up

Before you start using GridBlocks, you will need the Java Runtime Environment. Once you’ve installed that, start the Windows command prompt (Start > Run > “cmd”) and navigate to the folder you extracted the GB-DISK files to. To start the Front End service, use this command:

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java -jar gb-disk-fe-0.8.1.jar
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Once you’ve started up a Front End, you need to complement it with at least one Storage Element. You can start the SE service on the same PC as well, making it take part in the actual storage.